Semantic-World Knowledge Interactions During Processing

AIM: This ongoing study tracks real-time meaning formation using Visual World eye tracking (VWP) to test whether offline (Sentence-Picture Verification; SPV) data challenge the strict compositionality that many formal semantic theories assume.

BACKGROUND: The productivity and systematicity of language suggest linguistic meaning (semantics) is strictly compositional: the meaning of a complex expression is a context-insensitive function of the meanings of its parts (Dowty, 2007; Partee, 1995). But typicality effects caused by **lexical context dependence (LCD)** may pose a challenge: many rules (e.g., NP→Adj Noun) seem sensitive to world knowledge of what the units of composition refer to. For example, *red car, red hair,* and *red squirrel* all elicit different hues of red conceptually. LCD effects are found across various expression types in offline tasks (Lee, 2017; Matsuki et al., 2011; Poortman, 2017; Poortman et al., 2018; Ross et al., 2025; Zwaan et al., 2002). Our SPV data (**Fig 1a**) show LCD patterns at sentence-end. The question is *when* world knowledge is integrated with context-insensitive linguistic knowledge. We are now collecting VWP data to try to answer that question.

HYPOTHESES: (H1) Pre-Compositional Modification preserves strict compositionality by assuming broad semantic spans for each lexical item (e.g., *red* encompasses all reds) and context-driven referential narrowing (Del Pinal, 2016; cohort model of Marslen-Wilson & Welsh, 1978); **(H2) Modification During Composition** challenges strict compositionality by allowing lexical representations to mutually modify one another online in light of world knowledge (Pagin & Pelletier, 2007), producing the LCD pattern around noun offset + 400ms (NO+400) (based on semantic coercion data; e.g., Brennan & Pylkkänen, 2008; Piñango et al., 2006). We expect to recognize H2 both by LCD timing and by the presence, prior to NO+400, of lexical autonomy effects (Swinney (et al.), 1979, 2007; Tanenhaus et al, 1979); **(H3) Post-Compositional Modification** preserves strict compositionality by attributing LCD to post-compositional pragmatic inference (Del Pinal, 2016; Johnson & Keil, 2000), expected c. 1800ms after noun offset based on SPV results: Horchak & Garrido, 2021; Prystauka et al., 2023; Sato et al., 2013; Exp1 below.

EXP1: Prior SPV results suggest semantics interacts with object-state typicality. In our replication, subjects (n=288) read English sentences like *Oliver* VERB*ed the beach ball* then indicated whether an object (pictured 1000ms later) was mentioned in the sentence. Normed images depicted objects (beach ball) either in a typical (inflated) or atypical (deflated) state (**Fig. 1a**). We used no-change verbs (e.g., *notice*) to establish a baseline, and state-change verbs, the latter denoting change toward an object's typical (e.g., *inflate*) or atypical (e.g., *deflate*) state. We focus on the object-specific interactions between sentence and picture (**Fig 1b;** these are LCD effects evidenced in the noun encoding). The crucial comparison is between *inflate* and *deflate*. Generalized linear mixed effects models (outcome: response times (RTs)) confirm an interaction ($p<.001$**):** RTs were faster to typical noun states in the *inflate* than the *deflate* condition, and RTs were faster to atypical states in the *deflate* than the *inflate* condition. This result is a reference point for our test of real-time meaning computation using VWP in Exp 2 (data collection ongoing).

EXP2: To test the temporal predictions of H1-H3 in VWP, subjects ($n_{TARGET}=48$) hear, e.g., *Oliver* VERB*ed the beach ball* and view displays like **Fig. 2a**. Verbs (*notice, inflate, deflate*) match both objects (*beach ball, balloon*) to limit predictive processing. Planned Growth Curve Analyses contrasting *inflate/deflate* conditions will test *when,* during processing, fixations match SPV results (**Fig. 2b**): **(H1)** at the noun phrase (in line with strict compositionality); **(H2)** just after rapid semantic composition (against strict compositionality); **(H3)** well after compositional processing (in line with strict compositionality).

DISCUSSION: Theories of linguistic meaning have long debated whether meanings combine compositionally and how this process may be influenced by context and world knowledge. While SPV data provide rich information after meaning formation, VWP supports precise online tracking, potentially allowing us to determine whether LCD effects reflect non-compositional interaction in online linguistic structure building.

Figure 1. *Sentence-Picture Verification.*

(a) Stimuli (normed for typicality)

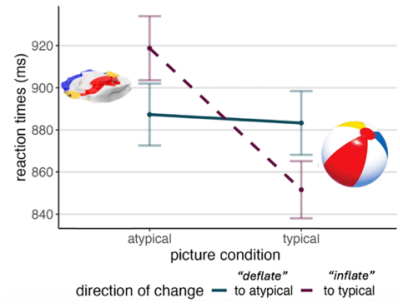
Images

atypical *typical*

Sentences

(a) Oliver <u>noticed</u> the beach ball. NO CHANGE
(b) Oliver <u>inflated</u> the beach ball. TO TYPICAL
(c) Oliver <u>deflated</u> the beach ball. TO ATYPICAL

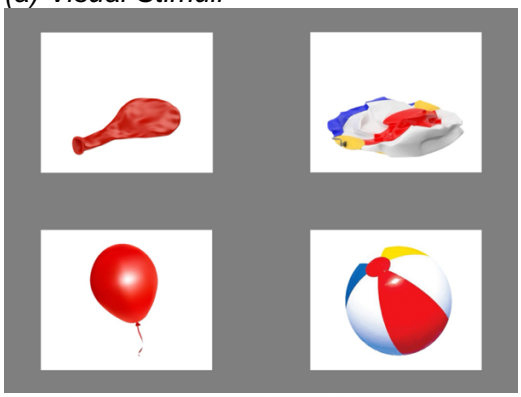
(b) Observed Mean Reaction Times and SEs



Note. Post-sentence RTs indicate an interaction between direction of change (lexically encoded in the verb) and object state typicality (a world knowledge property).

Figure 2. *Eye Tracking in the Visual World Paradigm.*

(a) Visual Stimuli



(b) Predictions for Each Hypothesis

At what point in the sentence will fixations match SPV results, indicating an LCD effect?

	Oliver	**VERB** *e.g., inflated*	**NOUN PHRASE** *e.g., the beach ball*	**AT 400MS AFTER NOUN OFFSET**	**AT 1800MS AFTER NOUN OFFSET**
H1			SPV		
H2				SPV	
H3					SPV

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